Juwon Lee

Education R&D · Developer, Seoul Robot & AI Museum

Target roles: Education planning & development

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Summary

At the Seoul Robot & AI Museum, I plan education programs and build the software used in classes and exhibits, from student web apps to exhibit kiosks and staff tools. My smart-farm program for the Yangpyeong Education Office was taught as a demo class to 5 high-school students in August 2026, ahead of a 15-student cohort in October, and my AI art exhibit generated 14,328 images on the museum floor from June to August 2026.

Experience

Education R&D · Developer · Seoul Robot & AI Museum (via operator FNJ Co., Ltd.) 2026.03 – Present

- Planned a 10-session, 20-hour AI and coding course for museum staff (one hour of theory and one hour of individual practice per session; five sessions are AI technology trend studies), scheduled for October 23 to December 23, 2026.
- Run the Picabot robot-arm drawing studio as a regular 30-minute hands-on AI session for visiting school groups; led its extended 50-minute class at the KT × Seoul City AI Future Camp (100 elementary students attended) in August 2026.
- Build internal tools behind exhibits and classes, including a staff scheduling app used daily by about 20 docents and part-time staff.

VQA Dataset Builder & QA Reviewer · Ubase, Inc. (dispatched to Select Star) 2025.11 – 2025.12

- Built and reviewed visual question-answering (VQA) datasets for LLM training on an ETRI (Electronics and Telecommunications Research Institute) project, including complex 3-hop reasoning Q&A sets.
- Sustained about 1.5× the team's average throughput while holding accuracy.

Youth Intern · Ministry of Science and ICT 2025.04 – 2025.08

- Supported improvements to the Pay-TV service quality evaluation and its satisfaction survey; drafted Broadcasting Act review materials.

Selected projects

Smart-Farm Education Program · Planning & development, solo 2026

IoT smart-farm and AI crop-diagnosis class commissioned by the Yangpyeong Education Office through the museum.

- Built the Next.js web app with server API routes on Vercel and Arduino (C++) firmware for the class's 15 ESP32 student boards (one per student), which write sensor readings straight to Supabase.

Mystery Playground RE:PLAY · Development, field tuning, operations tooling 2026

Ten-mission tablet game archiving the museum's 4th-floor special exhibition; five MediaPipe vision models run on the tablet.

- Traced misdrawn avatars to hats and headbands using the first 128 numbers-only visitor logs, then fixed the color, beard and curl rules and checked the change against a frozen eight-image benchmark.

Raimi's AI Art Lab · Planning, development, operations 2026

Guided prompt builder where visitors make AI artwork with the museum character Raimi.

- Replaced an off-the-shelf GPT service that froze mid-use with a custom web app and serverless backend; no freezes since the switch.

Exhibit Auto-Recovery · Diagnosis, build, rollout, handover 2026

Recovery system for a face- and voice-interaction robot exhibit (MaskBot) whose control software froze or exited mid-operation.

- Watchdog checks the program every 20 seconds and relaunches it within a minute of a crash or hang it can detect, with a two-stage check against false positives and a 5-minute pause after 3 restarts in a row.
- Recovery page with one button that any staff member can open on their own phone, reached over a Tailscale (WireGuard) mesh VPN with no port open to the internet.

Doctor-Green · 2024 prototype: team MAKENEW (PM & developer) · 2026 rebuild: solo 2024 – 2026

Smart-farm platform that puts IoT environment monitoring and control and crop-disease diagnosis on one web platform.

- Built the training set from AI Hub strawberry-disease data: 39,889 crop images in 5 classes (healthy plus 4 diseases) with a group-aware stratified split (train 32,155 / val 4,326 / test 3,408). Classifier training (ConvNeXt-Tiny or EfficientNetV2-S) is next; no results yet.

Teaching & mentoring

2026 SW·AI Donghaeng Project (KOFAC) · Youth mentor · makerthon coaching, live online class mentoring (middle school) 2026.09 – 2026.10

KB LA School (middle school) · Middle-school mentor · twice a week, 80 min · outside work hours 2026.07 – Present

Geuruteogi Learning Mentoring · University mentor (6th cohort) · one-on-one study mentoring for a Grade-9 student · twice a week, 90 min · outside work hours 2026.02 – Present

Skills & languages

Planning & Education: Product Planning (PM) · Learning Design · Bilingual Exhibition Guiding (KO·EN)
Frontend & App: Next.js · React · TypeScript · JavaScript · Tailwind · PWA · Capacitor
Backend, Infra & Ops: Supabase · Firebase / Firestore · Flask · Serverless Functions · Cloudflare R2 · Vercel · Tailscale / WireGuard
AI & Computer Vision: Python · TensorFlow.js · MobileNet v2 · MediaPipe · VGG16 / CNN · OpenAI API
Hardware · IoT · Robotics: ESP32 · MicroPython · Arduino / C++ · Sensors
Languages: Korean (native) · English (fluent; TOEIC Speaking AL, OPIc AL)

Awards

Prize · 2025 GH Youth Build-Up Start-up Competition (GH Corp.) 2026.03

Excellence Award (2nd place, finals) · 2025 Science Museum & Community AI Hackathon (National Science Museum) 2025.09

Excellence Award · ICT·SW Women's Start-up Competition 2024.12

Education

Hankyong National University 2022.03 – 2027.02

Plant Life & Environment, double major in Software Convergence · graduation requirements expected; graduation deferred